

# Online Repository: Age, co-morbidities and infections are associated with high postoperative visit frequency of chronic rhinosinusitis: a retrospective cluster study

## Authors

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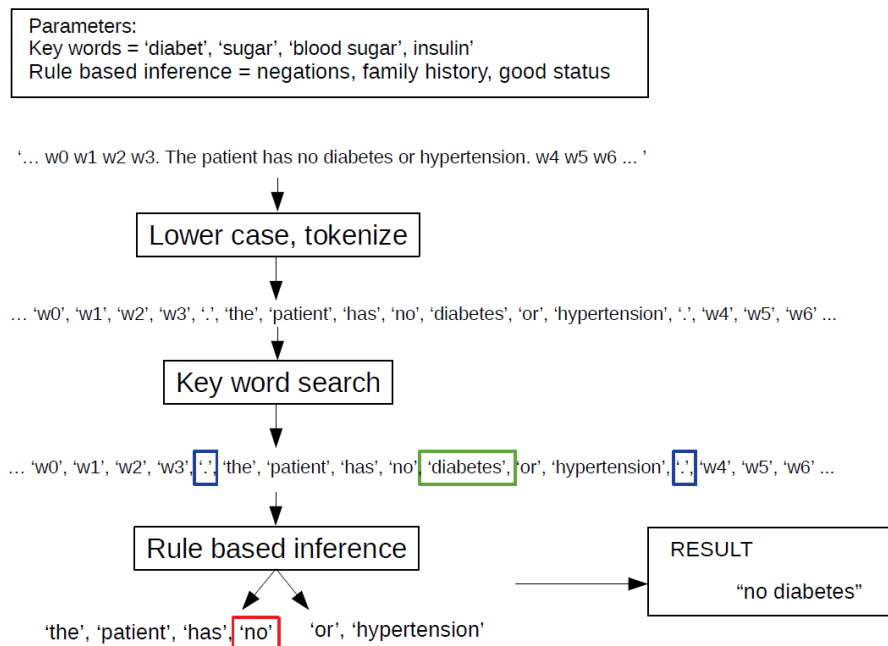
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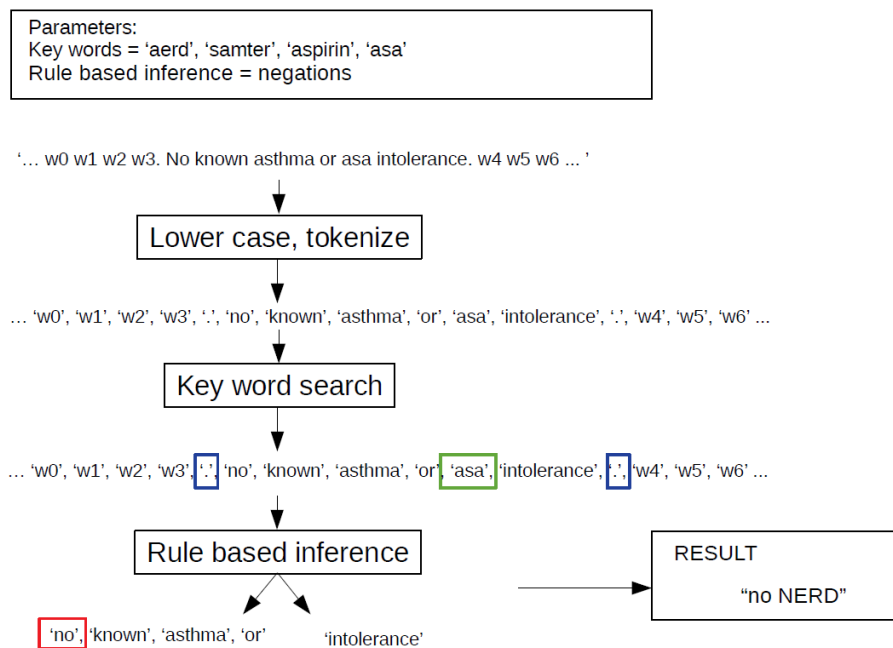
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**Table 1:** Key words for the mining of patient variables from the clinical text and disease specific dictionaries for evaluating when matched key words relate to the negation of disease, family history or good medical status

| Disease                         | Key word                                                                                                                                | Rule based dictionary                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Extra rules                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Allergy                         | cat<br>birch<br>prick<br>skin prick test                                                                                                | no, negative, mother, sister<br>no, negative, normal<br>no, negative, normal<br>no, negative, normal                                                                                                                                                                                                                                                                                                                                                                                        | Only "rast" or "rast-xxx" \\\                                                                                                           |
| Cancer                          | rast<br>malignant<br>carcinoma<br>basalioma<br>cancer<br>tumor<br>melanoma                                                              | no, negative<br>no<br>no<br>no<br>mother, father, family<br>no, benign                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                         |
| Cardiovascular disease          | hypertension<br>Coronary artery disease<br>coronary heart disease<br>aneurysm<br>cerebral hemorrhage<br>heart attack<br>angina<br>heart | no, normal, mother, father, family<br>no, normal, mother, father, family<br>no, normal, mother, father, family<br>no, normal, mother, father, family<br>no, normal, mother, father, family<br>no, normal, mother, father, family<br>no<br>normal, ok, ordinal, regular, good, balance,<br>family, mother, father, no, clean,<br>impeccable, clean, normal,<br>health, calm, vibrant, symmetric, parents<br>ordinal, okay, normal, compensation<br>regular, steady, calm, silent, no, family |                                                                                                                                         |
| Chronic respiratory disease     | rhythm<br>bronchitis<br>copd                                                                                                            | no, mother, father, family<br>no, mother, father, family                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                         |
| Diabetes                        | diabetes<br>insuline<br>sugar                                                                                                           | family, father, mother<br>normal, ok, father, mother, father, candied,<br>sugar free, food, fat, diet<br>good, normal, check                                                                                                                                                                                                                                                                                                                                                                | Same sentence should include: "disorder", \\<br>"problem", "difficulties", "regulation",<br>agressive", "control" or "self destructive" |
| Memory disorder                 | blood sugar<br>memory problem<br>memory impairment<br>memory disorder<br>Alzheimer<br>dementia                                          | no<br>no<br>caregiver, wife, husband, mother<br>caregiver, wife, husband, mother<br>work, working                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                         |
| Mental disorder                 | depression<br>mood disorder<br>adhd<br>psychiatrist<br>behavioral                                                                       | works, job, working                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                         |
| Musculoskeletal diseases        | osteoporosis<br>joint<br><br>tules<br>back                                                                                              | no, mother, father, family<br>no, normal, family, mother, father,<br>calm, good, working, over-moving<br><br>no, mother, father, family, close relative                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                         |
| Obesity                         | obesity                                                                                                                                 | no, mother, father, family                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Only "tules" or "tules-xxx"<br>Should be: back pain, back ache,<br>back problem, spinal cord injury,<br>ankylosing spondylitis          |
| NERD                            | fat<br>aerd<br>samter<br>aspirin<br>asa                                                                                                 | no, mother, father, family<br>no<br>no<br>no<br>no, mg                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                         |
| History of eosinophilia         | eosinofi                                                                                                                                | no, mild, slightly elevated, low, normal, negative                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                         |
| Smoking<br>Oral corticosteroids | smok<br>prednisolone<br>medrol<br>cortisone<br>corticosteroid<br>oral corticosteroid<br>oral cortisone                                  | no, mother, father, family<br>no<br><br><br><br><br><br>                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                         |



(a)



(b)

**Figure 1:** Examples of keyword search process of diabetes (a) and NERD (b) from clinical text. The keywords of the diabetes (translated to english) were: 'diabetes', 'sugar', 'blood sugar' and 'insulin'. The keywords for NERD were: 'aerd', 'samter', 'aspirin' and 'asa'. In the information extraction process, the input text is first lower cased and tokenized. The tokens from the clinical text are compared to the keywords. When the clinical text token and the keyword match (green squares), the tokens before and after the matched keyword are fed for the rule-based inference. The principle is that the selected tokens should belong in the same sentence (separated by blue squares) and the distance from the matched keyword should be smaller than 10 tokens. The rule-based inference uses disease specific dictionaries (Table E1) for evaluating when matched keywords relate to the negation of the disease, family history or good medical status.

**Table 2:** ICD-10 codes that were used for identifying chronic diseases.

| <b>Disease</b>             | <b>ICD-10 codes</b>             |
|----------------------------|---------------------------------|
| CRSsNP                     | J32 (without J33)               |
| CRSwNP                     | J33                             |
| Asthma                     | J45                             |
| Chronic pulmonary diseases | J40-J47, J80-J84, J90-94, G47.3 |
| Allergy                    | J30, J45.0                      |
| Obesity                    | E65-E67                         |
| Cardiovascular diseases    | I00-I99                         |
| Diabetes                   | E10-E14.8                       |
| Musculoskeletal diseases   | M00-M99                         |
| Cancer                     | C00-C97                         |
| Psychiatric diseases       | F00-F99                         |
| Memory diseases            | F00-F02, G30-, F11-F19          |